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Subject:: REVISION OF LMI/OGN/OPS/ELEC/004 ELECTRICAL TESTING OF GENERATOR ROTOR WINDING INSULATION.

Subject: REVISION OF LMI/OGN/OPS/ELEC/004 ELECTRICAL TESTING OF GENERATOR ROTOR WINDING INSULATION.

LMIs are reviewed periodically for further improvement and incorporation of Recommendation by internal or external audit committee and COS.

No changes envisaged by COS. Only periodic reviewing done after two years. No changes in the document.

May please approve.

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NTPC LIMITED
SIMHADRI SUPER THERMAL POWER STATION

LOCAL MANAGEMENT INSTRUCTION
LMI/OGN/OPS/ELEC/004
ISSUE NO.: 1, REVISION-4, DATE: July 2022

**ELECTRICAL TESTING OF GENERATOR ROTOR
WINDING INSULATION**

APPROVED FOR

IMPLEMENTATION by _____
HOP (SMTTP)

DATE _____

NTPC LTD.

SIMHADRI SUPER THERMAL POWER STATION

OPERATION AND MAINTENANCE DEPARTMENT

LMI(Name) :	ELECTRICAL TESTING OF GENERATOR ROTOR WINDING INSULATION		
LMI No.	LMI/OGN/OPS/ELEC/004		
ISSUE NO.	1	Revision : 4	Date :JULY 2022

	DESIGNATION	SIGNATURE & NAME
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ELECTRICAL TESTING OF GENERATOR ROTOR WINDING INSULATION

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1.0 OBJECTIVE:

The electrical insulation used in generator rotor windings has two functions. It provides the main insulation between the winding and the rotor body and between adjacent turns in the rotor winding.

Localized deterioration of the insulation in service will eventually result in the development of unacceptably low insulation resistance either to the rotor body (an earth fault) or between adjacent turns in the rotor windings (an inter-turn fault or shorted turn).

Although generators can operate with a single earth fault in the rotor winding, should a second earth fault develop catastrophic damage can result. It is of great importance, therefore, in deciding whether to continue in operation with an indicated earth fault, that its severity and position are accurately assessed.

Shorted turns develop in generator rotor windings when the inter-turn insulation between adjacent turns becomes defective. If the insulation resistance reduces significantly, current in the rotor winding may by-pass the affected turn. The operational significance of such current carrying inter-turn faults will be to increase the thermally dependent vibration of the rotor and/or cause overheating at the point of fault, which subsequently could result in a generator rotor earth fault.

In those cases where the inter-turn defect is of relatively high resistance and is not current carrying, it will have no immediate operational significance but may be indicative of long term deterioration of the insulation.

The aim of this LMI is to provide a standardized approach for monitoring the health of the electrical insulation systems of generator rotor windings during service life. This will allow the direct comparison of results obtained from manufacturers work tests and tests undertaken during operation of the generator.

2.0 SCOPE:

This Guidance Note recommends non-destructive electrical insulation test methods which are used in assessing the condition of the main and inter-turn insulation systems of the electrical windings of large generator rotors Unit 1, 2, 3 & 4..

3.0 REFERENCE DOCUMENT:

- a) COS-ISO-00-OGN/OPS/ELEC/004 Rev No.: 1 Date: June 2020 .
- b) Generator manual (O&M).

4.0 SUPERSEDED DOCUMENTS:

NIL

5.0 SPECIFIC INSTRUCTION TO BE FOLLOWED:

Table 1 gives guidance on the use of specific test methods available for detecting, locating and assessing the severity of rotor winding insulation faults.

Also indicated are the rotor conditions under which each test may be applied and the frequency at which it should be carried out.

6.0 METHODOLOGIES OF IMPLEMENTATION

ASSESSMENT OF INTER-TURN FAULTS

i) MEASUREMENT OF ROTOR WINDING IMPEDANCE

Development of a current-carrying inter turn fault reduces the overall impedance of the rotor winding. Applying an external A.C. voltage across the multi contact pins permits measurement of the rotor winding impedance at standstill of the rotor. refer annexure 2.

ii) Recurrent Surge Oscillograph (RSO) Method

In this test, low voltage, fast rise time pulses are injected into the rotor winding via the Multi pin contact. The preferred test waveform is a repetitive rectangular pulse. The pulses propagate through the

windings until they encounter a winding defect (shorted turn, earth fault, or open circuit) where a reflected wave is then generated. The shape and magnitude of the reflected wave is determined by the defect. The reflected wave propagates back to the injection point where it can be observed superimposed on the injected pulse. The time taken for the reflected pulse to return provides an indication of the position of the fault. Its magnitude and shape give an indication of the type and severity of the defect encountered.

This test can only be done with the rotor threaded out from the stator and cannot provide discrimination between current-carrying and non-current-carrying inter-turn faults.

III) Measurement of Rotor Winding Resistance

Development of a fault between adjacent turns of the rotor winding will reduce the overall winding resistance. A single completely short-circuited turn in a typical 500 MW generator rotor may reduce the overall resistance by approximately 1 per cent provided that rotor temperature maintained accurately. However, errors in the measurement of the rotor temperature will usually mask the effect of one shorted turn. The main use of this test is for detecting high resistance joints in the rotor winding. refer annexure 2.

IV) COMPARATIVE ASSESSMENT OF ROTOR EXCITATION CURRENT

A significant shorting in the rotor winding can be indicated by a sudden increase in the field excitation current for the same MW output. Hence the change in standard excitation / MW output relationship can be indicative of a rotor insulation fault. In addition, there may also be a sudden increase in the rotor bearing vibrations levels.

V) Measurement of Rotor Winding Insulation Resistance

Generator at stand still and Isolated from the Excitation System

The winding insulation resistance can be measured with the rotor stationary using a 500V electronic insulation tester.

Caution: The use of electronic instruments is preferred to avoid possible damage to diodes in the diode wheel system resulting from spikes in the output waveforms of hand-cranked instruments.

Note: The rotor earth fault alarm relay must be isolated and brush holder shall be removed. Faults within the main exciter rotor, diode wheel and multi contact pins will also be indicated by this test and cannot be separately identified.

VI) Measurement of D.C. Volt Drop along the Rotor Body (Rotor Withdrawn)

Axial location of a rotor winding earth fault can be made by passing a large direct current along the rotor body. The voltage between a winding multi contact pin and the rotor body is measured using a Milli voltmeter. The measurement connection to the rotor body is moved axially along the length of the rotor body until a nil indication is achieved on the meter. This indicates the approximate axial location of the winding defects. To pinpoint the second co-ordinate of the fault and hence locate it precisely, the test is repeated by passing the current across the rotor body in the transverse direction.

Caution: Care should be taken to establish an adequate connection for the test current to avoid damaging the rotor body. Adequately tightened current connections using bolt locations on rotor body must be used.

7.0 FREQUENCY OF CHECKING AND MEASUREMENTS:

Tests shall be carried out during generator major over hauling.

8.0 RESPONSIBILITY CENTERS FOR IMPLEMENTATION:

AGM Electrical maintenance - Generator group.

9.0 THE EXCEPTION REPORTING & DISPENSATION OF DEVIATIONS:

Any deviation shall be reported to HOD (EM).

10.0 RECORD KEEPING:

Records will be kept in respective LMI file (Gen & Excitation) Electrical Maintenance.

11.0 REVIEW:

Document will be reviewed after two years by HOP (Simhadri)

12.0 RECORDS OF REVISION:

AGM Electrical maintenance - Generator group.

ANNEXURE-1

Table 1 - Recommended Rotor Winding Tests

Text Reference		Generator Off-load & Stationary	Rotor Removed from Stator	Fault Location Capability
Tests for Inter turn Faults				
6.1	A.C. Impedance.	YES	YES	No
6.2	Recurrent Surge (RSO).	YES	YES	Yes
6.3	D.C resistance	YES	YES	No
Assessment of Rotor Fault in service				
6.4	Comparative Assessment Of Rotor Excitation Current	NO	NO	NO
Tests for locating earth Fault				
6.5	Insulation Resistance	YES	YES	No
6.6	D.C voltage drop	YES	YES	Yes

ANNEXURE-2

A. measurement of rotor winding impedance

sl no	applied voltage	measured current	calculated impedance

B. Measurement of rotor winding resistance

[illegible]